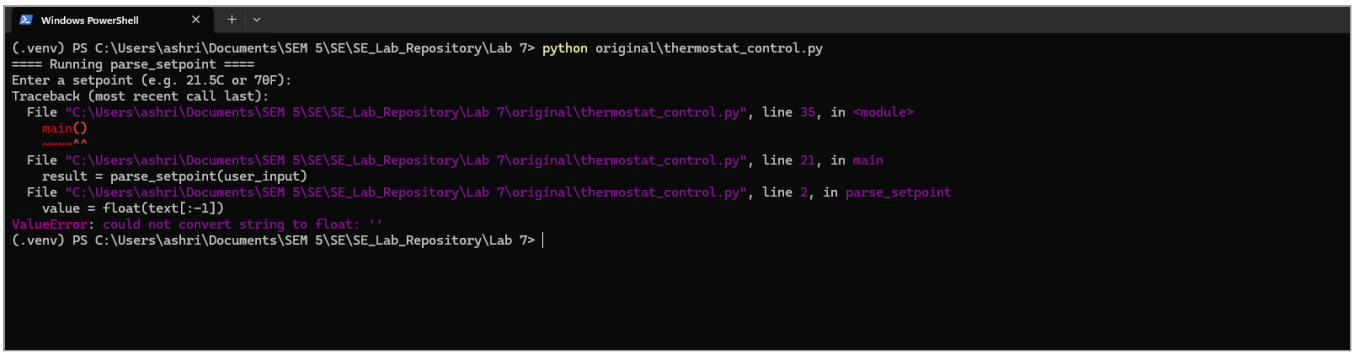
Lab 7 – Fuzzing for Input Validation Bugs

Ashrit Nair · PES2UG24AM032 · Assigned repo `SETAPESU26/L6_L7_29_thermostat_control` (lab7) · Tool: Hypothesis 6.170 + pytest

Step 2 – Running the original application

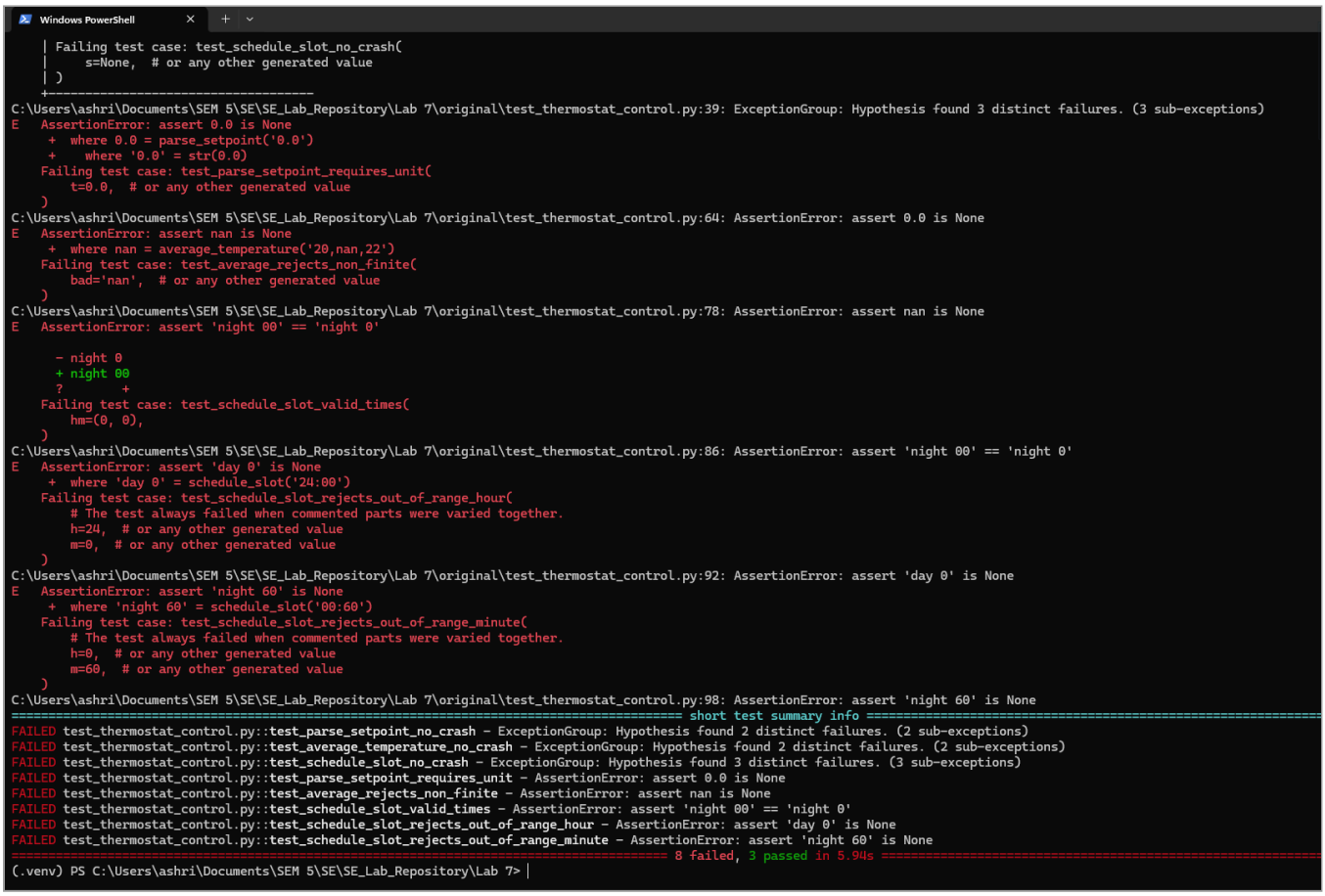
An empty setpoint crashes the program immediately with an unhandled `ValueError`.



Step 3 – Hypothesis tests written

`test_thermostat_control.py` has 11 tests at 500 generated examples each. Three are no-crash fuzzers (`st.text() | st.none()`) for each function. Eight are property tests: a Celsius/Fahrenheit round trip, a mandatory unit, the average staying within min..max, non-finite readings rejected, valid HH:MM formatting, and out-of-range hours and minutes rejected.

Step 4 – Fuzz tests against the original code: 8 failed, 3 passed

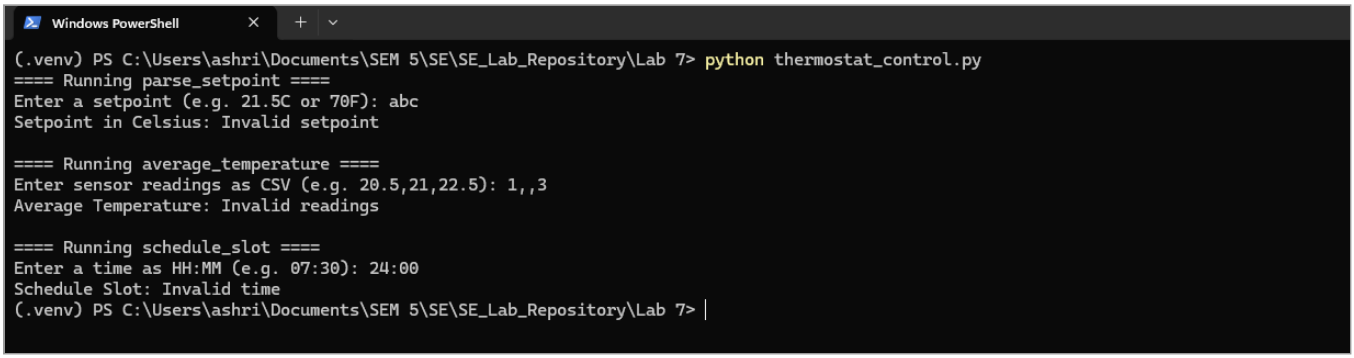


Bugs uncovered and fixes

Function	Minimal input found	Original behaviour	Fix
parse_setpoint	None	TypeError	type check → None
parse_setpoint	`''`	ValueError	regex: number + mandatory C/F unit
parse_setpoint	`'0.0'`	silently 0.0 (read '0.' with unit '0')	unit required → None
parse_setpoint	`'1e400C'`	returned inf	reject non-finite
average_temperature	None	AttributeError	type check → None
average_temperature	`'', '1,,3'`	ValueError	per-value try/except → None
average_temperature	`'20,nan,22'`	returned nan	reject non-finite readings
schedule_slot	None	AttributeError	type check → None
schedule_slot	`'', ':'`	ValueError	regex H:MM / HH:MM
schedule_slot	`'24:00', '00:60'`	accepted	range check 0–23 / 0–59
schedule_slot	`'00:00'`	'night 00' (but daytime gives 'day 0')	normalise minute with int()

Design choice: every function now returns `None` for invalid input instead of raising, and `main()` prints a clear "Invalid …" message. Valid inputs give the same results as before.

Step 5 – Running the fixed application with the same kinds of bad input



Step 6 – Fuzz tests against the fixed code: 11 passed

